ic_autolab round 11 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T04:37:51Z at git `2c0d54ecc8eac10981bfa1d15e8cf3245942354c`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-f74cbc` (`python3 tools/pcc/pcc.py standing PCC-20261104-f74cbc`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
24	**no_challenger**	none (best candidate `ksum4-n10`, train S 6.959 / 7.815 / 15.519)	-	-	-	-	-	6.607 / 7.064 / 11.171	1.147 / 1.250 / 2.177	inc: 5.8x / 5.7x / 5.1x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

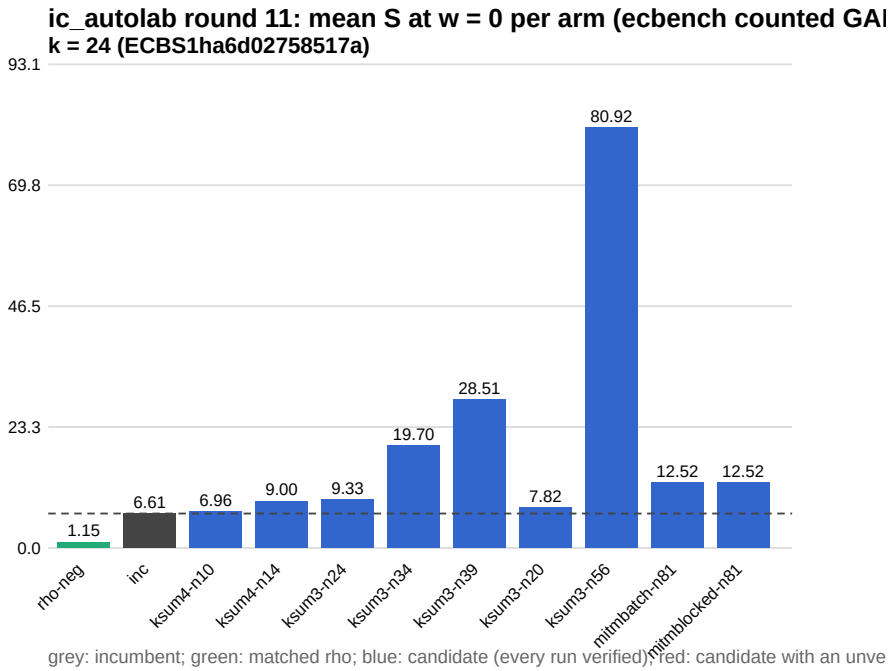
- k = 24: no_challenger

Train (development screen, in sample)

k = 24: session `ECBS1ha6d02758517a` (run exit 0, verify exit 0, receipt sha256 `30dc04a1043b1f275910beb71735126c68a8e4cfb6e01b79101c375b5b9af890`)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	32	32	1.147 / 1.250 / 2.177	matched reference
`inc`	ksum4	12	`prime-projected-abscissa:size=12,scan_cap=192`	`ksum:m=4`	32	32	6.607 / 7.064 / 11.171	incumbent
`ksum4-n10`	ksum4	10	`prime-projected-abscissa:size=10,scan_cap=160`	`ksum:m=4`	32	32	6.959 / 7.815 / 15.519	incumbent neighbourhood: N = 0.85 x 12

arm	family	N factor base	oracle	runs verified	S (w = 0 / 0.1 / 1)	reason proposed
ksum4-n14	ksum4	14 prime-projected-abscissa:size=14,scan_cap=224	ksum:m=4	32	32	9.003 / 9.308 / 12.050	incumbent neighbourhood: N = 1.2 x 12
ksum3-n24	ksum3	24 prime-projected-abscissa:size=24,scan_cap=384	ksum:m=3	32	32	9.334 / 9.812 / 14.117	cross-size transfer neighbourhood: N = 0.85 x 28 (ksum3 from k = 20)
ksum3-n34	ksum3	34 prime-projected-abscissa:size=34,scan_cap=544	ksum:m=3	32	32	19.705 / 20.012 / 22.772	cross-size transfer neighbourhood: N = 1.2 x 28 (ksum3 from k = 20)
ksum3-n39	ksum3	39 prime-projected-abscissa:size=39,scan_cap=624	ksum:m=3	32	32	28.511 / 28.784 / 31.242	cross-size transfer neighbourhood: N = 1.4 x 28 (ksum3 from k = 20)
ksum3-n20	ksum3	20 prime-projected-abscissa:size=20,scan_cap=320	ksum:m=3	32	32	7.824 / 8.494 / 14.517	cross-size transfer neighbourhood: N = 0.7 x 28 (ksum3 from k = 20)
ksum3-n56	ksum3	56 prime-projected-abscissa:size=56,scan_cap=896	ksum:m=3	32	32	80.916 / 81.255 / 84.302	cross-size transfer neighbourhood: N = 2 x 28 (ksum3 from k = 20)
mitmbatch-n81	mitmbatch	81 prime-projected-abscissa:size=81,scan_cap=1296	mitm-prime-batch:m=2	32	32	12.517 / 13.628 / 23.626	exploration: family mitmbatch untried at k = 24, N = 1 x N_a
mitmblocked-n81	mitmblocked	81 prime-projected-abscissa:size=81,scan_cap=1296	mitm-prime-blocked:m=2	32	32	12.517 / 13.628 / 23.626	exploration: family mitmblocked untried at k = 24, N = 1 x N_a



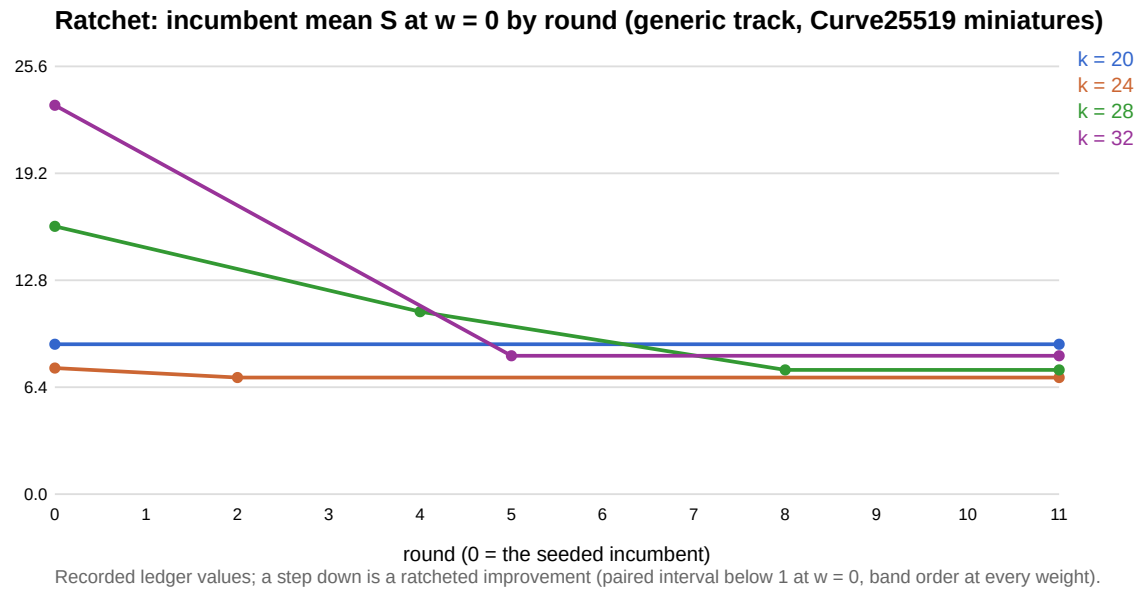
train screen

Selection

- k = 24: no_challenger (Pareto alternative at w = 1: `ksum4-n14`)

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`ksum3-n11` (`prime-projected-abscissa:size=11,scan_cap=176` + `ksum:m=3`)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	`ksum4-n12` (`prime-projected-abscissa:size=12,scan_cap=192` + `ksum:m=4`)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	`ksum4-n18` (`prime-projected-abscissa:size=18,scan_cap=288` + `ksum:m=4`)	8	7.445 / 7.910 / 12.100	7.8x / 7.6x / 6.5x
32	`ksum4-n26` (`prime-projected-abscissa:size=26,scan_cap=416` + `ksum:m=4`)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.